

Advanced Topics in Deep Learning
Imitation Learning for LunarLander-v3
Assignment Report

Submitted as mid-semester project report for Advanced Topics in Deep Learning,
Colman, 2026

Contents

1	Introduction	3
1.1	Environment and Task Description	3
1.2	Assignment Pipeline Overview	3
2	Stage A: Expert Reinforcement Learning Agent	3
2.1	Provided PPO Notebook	3
2.2	Training Setup and Hyperparameters	3
2.3	Stopping Criteria	4
2.4	Expert Performance Evaluation	4
3	Stage B: Demonstration Data Collection	4
3.1	Data Collection Procedure	4
3.2	Dataset Sizes	4
3.3	Pseudo-code: Demonstration Collection	5
4	Stage C: Imitation Learning (Behavioral Cloning)	5
4.1	Model Architecture	5
4.2	Training Procedure	5
4.3	Pseudo-code: Behavioral Cloning	5
4.4	Training Curves	6
5	Stage D: Evaluation of the Imitation Policy	6
5.1	Evaluation Metrics	6
5.2	Expert vs Imitation Comparison	7
5.3	Video Results	7
6	Stage E: Analysis of Data Quantity	7
6.1	Performance vs Dataset Size	7
6.2	Discussion	8
7	Competitive Section: Few-Shot Imitation Learning	8
7.1	Few-Shot Setup	8
7.2	Results	8
7.3	Insights: Data Quantity vs Data Quality	8
8	Conclusion	9
9	Links to Code and Artifacts	9

1 Introduction

1.1 Environment and Task Description

The LunarLander-v3 environment is a classic continuous-control-inspired task with a discrete action space. The agent observes an 8-dimensional state vector describing the lander’s position, velocity, angle, angular velocity, and contact sensors. The action space consists of four discrete actions: no-op, fire left engine, fire main engine, and fire right engine.

The objective of the task is to land the spacecraft safely between the flags with minimal fuel consumption. An episode is considered successful when the total reward exceeds 200, following the standard benchmark used in prior work and the provided expert notebook.

1.2 Assignment Pipeline Overview

The project follows a five-stage pipeline:

- **Stage A:** Training and evaluation of an expert reinforcement learning agent (provided by the instructor).
- **Stage B:** Collection of expert demonstrations.
- **Stage C:** Training imitation policies via Behavioral Cloning.
- **Stage D:** Quantitative and qualitative evaluation of imitation agents.
- **Stage E:** Analysis of the impact of the size of the demonstration dataset.

Only Stages B–E were implemented by us. Stage A is analyzed and reported without claiming authorship.

2 Stage A: Expert Reinforcement Learning Agent

2.1 Provided PPO Notebook

The expert agent was trained using Proximal Policy Optimization (PPO) with an MLP policy, as provided by the course staff. The notebook includes full training logs, evaluation curves, and video recordings. The trained expert is treated as a fixed oracle and is not modified in this project.

2.2 Training Setup and Hyperparameters

Key hyperparameters include:

- Algorithm: PPO (Stable-Baselines3)
- Policy: MLP
- Parallel environments: 8
- Learning rate: 3×10^{-4}
- Discount factor $\gamma = 0.99$
- Total timesteps: 1.2×10^6

2.3 Stopping Criteria

Training is stopped after a fixed budget of 1.2M environment steps. Additionally, an evaluation callback monitors performance every 25,000 steps over 50 episodes, saving the best-performing model.

2.4 Expert Performance Evaluation

The final expert achieves a mean reward of approximately 269–271 with a standard deviation of about 18–19 over 50 evaluation episodes. This exceeds the success threshold of 200, and the environment is considered solved. This expert policy serves as the source of demonstrations for all subsequent stages.

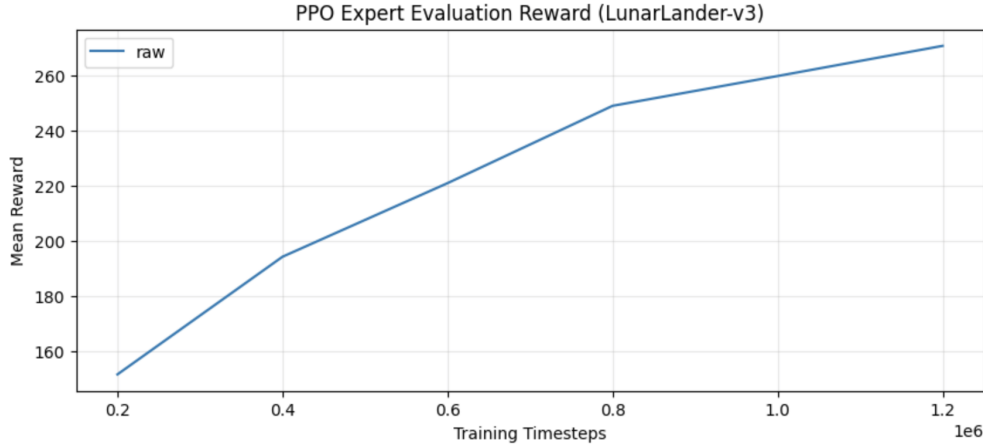


Figure 1: PPO Expert Evaluation Reward over training timesteps. The agent steadily improves, crossing the solved threshold (> 200).

3 Stage B: Demonstration Data Collection

3.1 Data Collection Procedure

Demonstrations are collected by running the expert policy deterministically and recording state-action pairs at every environment step. Multiple datasets of increasing size are generated by varying the number of expert episodes.

3.2 Dataset Sizes

The following datasets were collected:

- 3 episodes: 726 steps
- 5 episodes: 1,262 steps
- 10 episodes: 3,209 steps
- 50 episodes: 13,350 steps
- 100 episodes: 26,860 steps
- 250 episodes: 68,607 steps
- 500 episodes: 136,934 steps

The average rewards for experts during data collection remain consistently high ($\approx 265\text{--}285$), indicating stable expert behavior.

3.3 Pseudo-code: Demonstration Collection

Algorithm 1 Expert Demonstration Collection

```

for each episode do
  reset environment
  while episode not terminated do
    record current state
    select action using expert policy
    execute action in environment
    record action
  end while
end for
save all state-action pairs to disk

```

4 Stage C: Imitation Learning (Behavioral Cloning)

4.1 Model Architecture

All imitation models share the same architecture:

- Input: 8-dimensional state vector
- Two fully connected layers with 64 ReLU units each
- Output: 4-way softmax over discrete actions

The total number of trainable parameters is 4,996.

4.2 Training Procedure

Actions are converted to one-hot vectors and optimized using categorical cross-entropy loss. Training is performed for 30 epochs with batch size 64 and an 80/20 train-validation split.

4.3 Pseudo-code: Behavioral Cloning

Algorithm 2 Behavioral Cloning

```

initialize neural network policy  $\pi_\theta$ 
for each training epoch do
  for each minibatch of demonstrations do
    predict action probabilities
    compute cross-entropy loss
    update  $\theta$  via gradient descent
  end for
end for

```

4.4 Training Curves

For the 100-episode dataset, the training converges smoothly. The validation accuracy increases from approximately 67% to approximately 95.9%, while the validation loss steadily decreases. No significant overfitting is observed.

Below, we present the training curves for the Large dataset (100 episodes) followed by the Small dataset (10 episodes) for comparison.

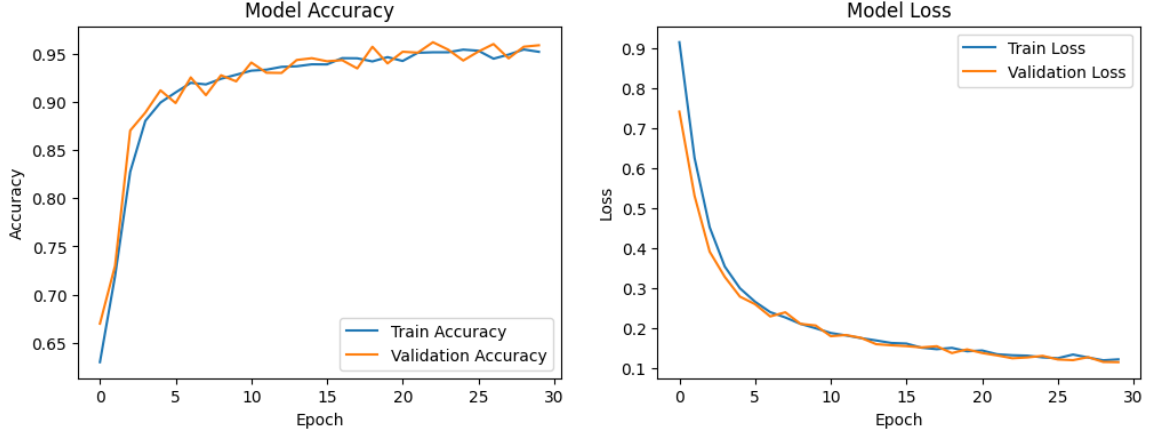


Figure 2: Training curves for the 100-episode (Large) dataset. Note the smooth convergence and high final accuracy ($\approx 96\%$).

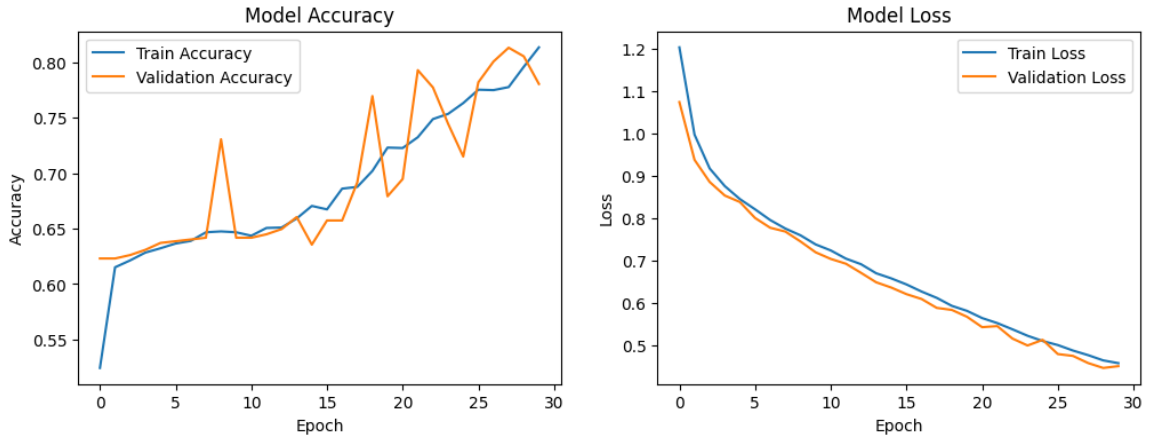


Figure 3: Training curves for the Small dataset (Few-Shot). Compared to the Large dataset, the accuracy is lower ($\approx 80\%$) and the curves exhibit more noise.

5 Stage D: Evaluation of the Imitation Policy

5.1 Evaluation Metrics

Imitation policies are evaluated over 100 episodes using:

- Success rate (episodes with reward ≥ 200)
- Average episode reward

5.2 Expert vs Imitation Comparison

Performance improves sharply with the size of the data set. Key results include:

- 3 episodes: 0% success, average reward -182
- 5 episodes: 75% success, average reward ≈ 189
- 10 episodes: 94% success, average reward ≈ 258
- 50+ episodes: 96–98% success, average reward ≈ 264 –267

Beyond 50 episodes, performance saturates and matches closely with the expert.

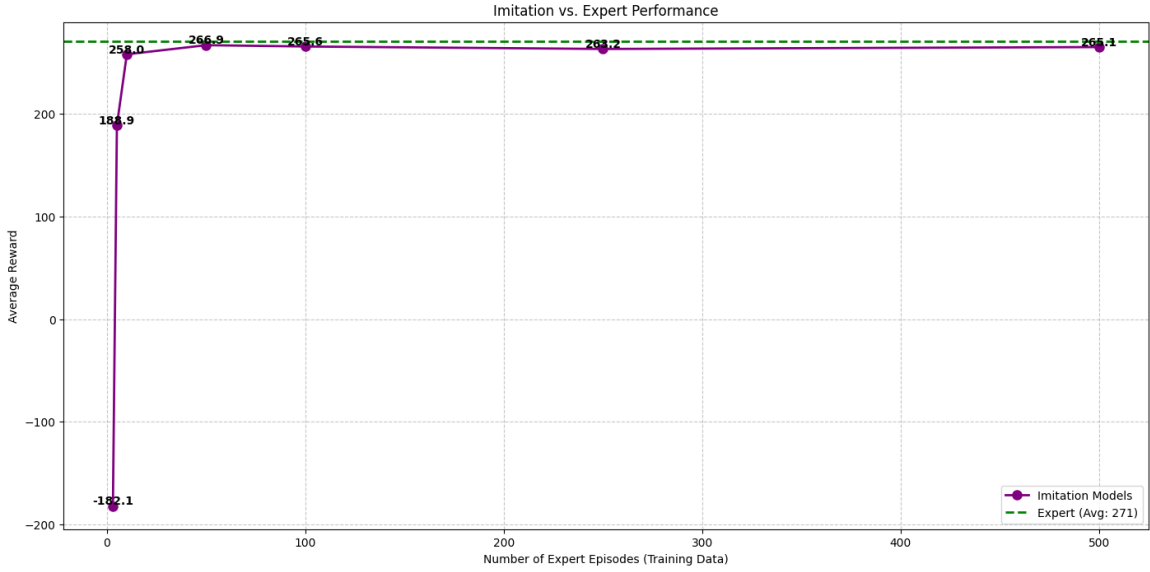


Figure 4: Imitation vs. Expert Performance (Average Reward). The purple line represents the imitation models, while the green dashed line represents the expert baseline. Saturation occurs around 10-50 episodes.

5.3 Video Results

Inference videos demonstrate stable landings and smooth trajectories. The selected 10-episode model achieves a mean reward of 264.9 with low variance during the inference-only evaluation, which confirms generalization. ([Videos](#))

6 Stage E: Analysis of Data Quantity

6.1 Performance vs Dataset Size

Both the success rate and the average reward exhibit diminishing returns. The greatest improvement occurs between 3 and 10 episodes. Beyond approximately 50 episodes, additional data yields marginal gains.

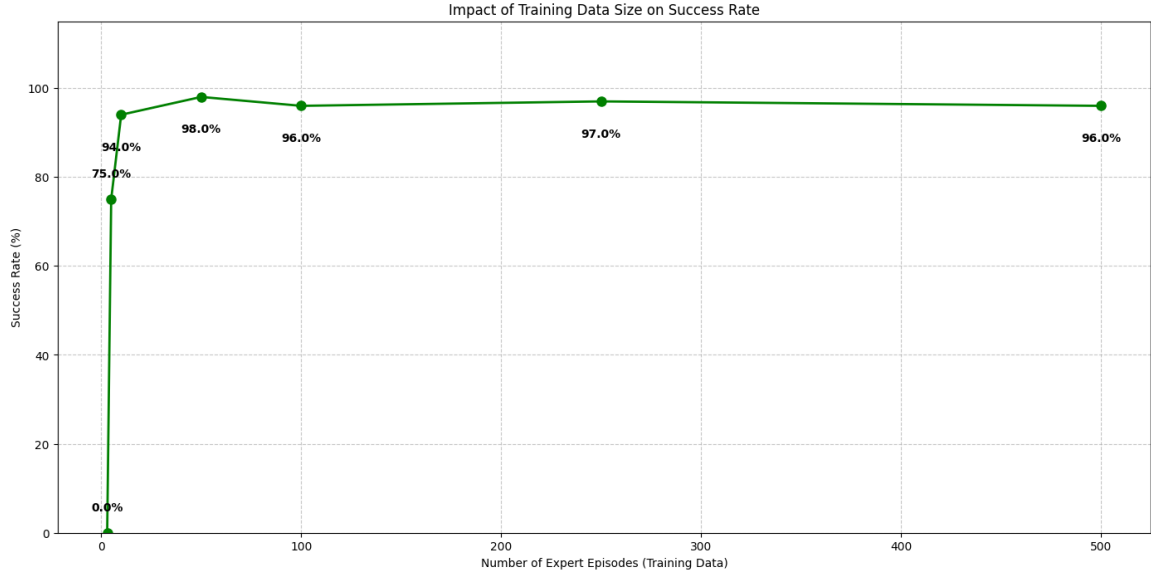


Figure 5: Impact of Training Data Size on Success Rate. The model reaches $> 90\%$ success rate with as few as 10 episodes.

6.2 Discussion

The results indicate a clear saturation point. While very small datasets fail catastrophically, a relatively modest number of high-quality expert demonstrations is sufficient to recover near-expert performance.

7 Competitive Section: Few-Shot Imitation Learning

7.1 Few-Shot Setup

Few-shot imitation is evaluated using datasets of 3 and 5 expert episodes, without any algorithmic modification beyond standard Behavioral Cloning.

7.2 Results

The 3-episode model completely fails. The 5-episode model partially succeeds, achieving a 75% success rate but with significantly reduced reward compared to the expert. When comparing the training process, the small dataset shows significantly higher variance and lower final accuracy, as seen in Figure 3 in Section 4.4 compared to the large dataset.

7.3 Insights: Data Quantity vs Data Quality

Based strictly on the observed results, data quality is a prerequisite, but data quantity determines reliability. High-quality expert behavior is necessary but insufficient when the dataset is too small. Once a minimal coverage threshold is reached (around 10–50 episodes), additional data provides limited benefit.

Conclusion: After a small critical mass, data quality matters more than sheer quantity, but below that threshold, quantity is decisive.

8 Conclusion

This project demonstrates that Behavioral Cloning can successfully replicate a strong PPO expert on LunarLander-v3. With surprisingly few demonstrations, imitation policies approach expert-level performance. However, the results also highlight the limitations of naive imitation in the extreme few-shot regime.

9 Links to Code and Artifacts

- [Stage A \(RL Expert\)](#)
- [Stage B-E \(Imitation Learning\)](#)
- [Inference](#)
- [Videos](#)